AERIS Expert Review Packet

Anomaly

Source / metric	openaq / PM2.5 (pm25)
Observed / expected baseline	62.2 / 8.74196 ug/m3
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-06-07 07:00 UTC
Location	29.8145, -95.3877
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 640	59.7 to 100; mean 84.9022	100 at 2026-06-07 06:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 640	23.8889 to 32.7778; mean 27.3693	26 at 2026-06-07 06:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 640	0 to 360; mean 141.719	170 at 2026-06-07 06:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 640	0 to 10.8033; mean 3.6558	5.65888 at 2026-06-07 06:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	11 / 132	5853.96 to 5909.97; mean 5873.6	5867.9 at 2026-06-07 06:00Z; 1 h before event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	11 / 132	44.2979 to 1876.79; mean 646.683	520.634 at 2026-06-07 06:00Z; 1 h before event
noaa_gfs	Precipitable water (precipitable_water)	mm	11 / 132	49.7819 to 59.8974; mean 55.2693	54.0192 at 2026-06-07 06:00Z; 1 h before event
noaa_gfs	Surface pressure (surface_pressure)	hPa	11 / 132	1001.91 to 1014.2; mean 1007.98	1007.01 at 2026-06-07 06:00Z; 1 h before event
noaa_gfs	850-hPa temperature (t_850)	degC	11 / 132	16.5548 to 20.0187; mean 18.6141	19.2674 at 2026-06-07 06:00Z; 1 h before event
noaa_gfs	10-m eastward wind (u_10m)	m/s	11 / 132	-4.29892 to 1.30517; mean -1.2014	-1.34029 at 2026-06-07 06:00Z; 1 h before event
noaa_gfs	10-m northward wind (v_10m)	m/s	11 / 132	0.345771 to 6.44698; mean 3.1582	3.15933 at 2026-06-07 06:00Z; 1 h before event
openaq	Ozone (ozone)	ppm	18 / 1234	0 to 0.038; mean 0.0169	0.011 at 2026-06-07 07:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 207	6 to 87; mean 33.1594	61 at 2026-06-07 07:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	88 / 14157	-4.9 to 177.77; mean 6.9481	62.2 at 2026-06-07 07:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 360	76 to 100; mean 98.4944	100 at 2026-06-07 06:57Z; 2 min before event
openweather	Relative humidity (humidity)	percent	5 / 360	67 to 96; mean 86.2778	93 at 2026-06-07 06:57Z; 2 min before event
openweather	Precipitation rate (precipitation)	mm/h	5 / 360	0 to 7.49; mean 0.0532	0 at 2026-06-07 06:57Z; 2 min before event
openweather	Surface pressure (pressure)	hPa	5 / 360	1008 to 1015; mean 1010.9	1010 at 2026-06-07 06:57Z; 2 min before event
openweather	Air temperature (temperature)	degC	5 / 360	24.44 to 32.69; mean 27.5534	26.44 at 2026-06-07 06:57Z; 2 min before event
openweather	Wind direction (wind_direction)	degree	5 / 360	0 to 355; mean 152.492	175 at 2026-06-07 06:57Z; 2 min before event
openweather	Wind speed (wind_speed)	m/s	5 / 360	0.45 to 5.94; mean 2.2147	1.79 at 2026-06-07 06:57Z; 2 min before event
purpleair	PM2.5 (pm25)	ug/m3	66 / 4771	0 to 286.4; mean 6.5099	8.7 at 2026-06-07 07:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-06-06 20:14Z; 10 h 45 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-06-06 20:14Z; 10 h 45 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	2 / 2	0.0279382 to 0.0279482; mean 0.0279	0.0279482 at 2026-06-06 20:14Z; 10 h 45 min before event
sentinel5p	CO granules available (s5p_co_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-06-06 20:14Z; 10 h 45 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	2 / 2	0.000118207 to 0.00012874; mean 0.0001	0.00012874 at 2026-06-06 20:14Z; 10 h 45 min before event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-06-06 20:14Z; 10 h 45 min before event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	1 / 1	3.15424e-05 to 3.15424e-05; mean 0	3.15424e-05 at 2026-06-06 20:14Z; 10 h 45 min before event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-06 20:14Z; 10 h 45 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-06-06 20:14Z; 10 h 45 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	2 / 2	-9.7923e-05 to -9.174e-05; mean -0.0001	-9.7923e-05 at 2026-06-06 20:14Z; 10 h 45 min before event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-06-06 20:14Z; 10 h 45 min before event
tceq	Carbon monoxide (co)	ppm	3 / 215	0.1 to 0.8; mean 0.2247	0.2 at 2026-06-07 07:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	13 / 908	-2.7 to 28.3; mean 3.4452	2 at 2026-06-07 07:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 211	-0.5 to 0.7; mean -0.145	-0.2 at 2026-06-07 07:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	06-05 18Z 81, 06-06 00Z 86.41, 06Z 93.76, 12Z 83.45, 18Z 82.93, 06-07 00Z 90.42, 06Z 92.7, 12Z 78.11, 18Z 73.18, 06-08 00Z 87.43, 06Z 91.28, 12Z 80.87, 18Z 65.98
asos	Air temperature (temperature)	06-05 18Z 28.02, 06-06 00Z 26.2, 06Z 24.98, 12Z 27.9, 18Z 27.62, 06-07 00Z 27.02, 06Z 26.41, 12Z 28.25, 18Z 28.63, 06-08 00Z 27.13, 06Z 26.81, 12Z 28.79, 18Z 31.43
asos	Wind direction (wind_direction)	06-05 18Z 122.7, 06-06 00Z 109.4, 06Z 83.14, 12Z 131.5, 18Z 154.2, 06-07 00Z 150.4, 06Z 146.7, 12Z 162.5, 18Z 164.8, 06-08 00Z 145.4, 06Z 158.3, 12Z 161.9, 18Z 165.6
asos	Wind speed (wind_speed)	06-05 18Z 4.676, 06-06 00Z 3.164, 06Z 1.574, 12Z 2.267, 18Z 3.294, 06-07 00Z 3.468, 06Z 3.02, 12Z 5.026, 18Z 5.011, 06-08 00Z 3.639, 06Z 3.915, 12Z 4.697, 18Z 4.859
noaa_gfs	500-hPa geopotential height (gh_500)	06-06 00Z 5865, 06Z 5864, 12Z 5857, 18Z 5870, 06-07 00Z 5864, 06Z 5868, 12Z 5862, 18Z 5873, 06-08 00Z 5873, 06Z 5889, 12Z 5890, 18Z 5909
noaa_gfs	Planetary boundary layer height (pbl_height)	06-06 00Z 687.4, 06Z 239.4, 12Z 116.6, 18Z 1099, 06-07 00Z 402.7, 06Z 431.6, 12Z 477.5, 18Z 1193, 06-08 00Z 612.9, 06Z 423.5, 12Z 509.8, 18Z 1567
noaa_gfs	Precipitable water (precipitable_water)	06-06 00Z 52.65, 06Z 54.45, 12Z 55.58, 18Z 57.27, 06-07 00Z 58.09, 06Z 54.47, 12Z 56.13, 18Z 53.89, 06-08 00Z 53.94, 06Z 57.45, 12Z 55.9, 18Z 53.42
noaa_gfs	Surface pressure (surface_pressure)	06-06 00Z 1007, 06Z 1008, 12Z 1008, 18Z 1008, 06-07 00Z 1006, 06Z 1007, 12Z 1007, 18Z 1008, 06-08 00Z 1007, 06Z 1009, 12Z 1010, 18Z 1012
noaa_gfs	850-hPa temperature (t_850)	06-06 00Z 18.13, 06Z 17.94, 12Z 17.73, 18Z 18.3, 06-07 00Z 19.08, 06Z 19.21, 12Z 18.47, 18Z 19.08, 06-08 00Z 19.47, 06Z 19.57, 12Z 19.25, 18Z 17.13
noaa_gfs	10-m eastward wind (u_10m)	06-06 00Z -3.555, 06Z -1.958, 12Z -0.7138, 18Z 0.7479, 06-07 00Z -1.034, 06Z -1.048, 12Z -0.7821, 18Z -1.072, 06-08 00Z -1.02, 06Z -1.241, 12Z -1.224, 18Z -1.518
noaa_gfs	10-m northward wind (v_10m)	06-06 00Z 2.316, 06Z 1.347, 12Z 1.509, 18Z 2.598, 06-07 00Z 3.312, 06Z 3.188, 12Z 3.295, 18Z 5.437, 06-08 00Z 3.679, 06Z 2.678, 12Z 3.57, 18Z 4.972
openaq	Ozone (ozone)	06-05 18Z 0.02577, 06-06 00Z 0.01893, 06Z 0.01189, 12Z 0.01376, 18Z 0.02306, 06-07 00Z 0.01591, 06Z 0.01245, 12Z 0.01811, 18Z 0.02124, 06-08 00Z 0.01469, 06Z 0.01534, 12Z 0.01342, 18Z 0.02224
openaq	PM10 (pm10)	06-05 18Z 12.33, 06-06 00Z 20.08, 06Z 13.44, 12Z 19.39, 18Z 13.17, 06-07 00Z 40.83, 06Z 51.06, 12Z 43.61, 18Z 41.22, 06-08 00Z 40.94, 06Z 39.5, 12Z 43.5, 18Z 45.33
openaq	PM2.5 (pm25)	06-05 18Z 3.917, 06-06 00Z 2.954, 06Z 3.064, 12Z 4.436, 18Z 4.081, 06-07 00Z 7.865, 06Z 9.41, 12Z 7.992, 18Z 6.984, 06-08 00Z 8.275, 06Z 8.576, 12Z 8.057, 18Z 6.731
openweather	Cloud cover (cloud_cover)	06-05 18Z 99.48, 06-06 00Z 98.83, 06Z 100, 12Z 99.9, 18Z 94.5, 06-07 00Z 93.43, 06Z 98.77, 12Z 97.8, 18Z 100, 06-08 00Z 99.53, 06Z 99.93, 12Z 100, 18Z 98
openweather	Relative humidity (humidity)	06-05 18Z 84.76, 06-06 00Z 87.57, 06Z 92.7, 12Z 85.87, 18Z 82.93, 06-07 00Z 90.23, 06Z 92.57, 12Z 82.47, 18Z 77, 06-08 00Z 87.13, 06Z 91.3, 12Z 82.97, 18Z 71.8
openweather	Precipitation rate (precipitation)	06-05 18Z 0.0692, 06-06 00Z 0, 06Z 0, 12Z 0.1037, 18Z 0.3413, 06-07 00Z 0, 06Z 0, 12Z 0, 18Z 0.003667, 06-08 00Z 0, 06Z 0.08733, 12Z 0.045, 18Z 0
openweather	Surface pressure (pressure)	06-05 18Z 1011, 06-06 00Z 1011, 06Z 1010, 12Z 1011, 18Z 1009, 06-07 00Z 1010, 06Z 1009, 12Z 1011, 18Z 1010, 06-08 00Z 1011, 06Z 1012, 12Z 1014, 18Z 1014
openweather	Air temperature (temperature)	06-05 18Z 28.19, 06-06 00Z 26.3, 06Z 24.96, 12Z 27.96, 18Z 28.49, 06-07 00Z 27.14, 06Z 26.34, 12Z 28.42, 18Z 28.97, 06-08 00Z 27.36, 06Z 26.87, 12Z 29.01, 18Z 32.03
openweather	Wind direction (wind_direction)	06-05 18Z 118.3, 06-06 00Z 116.8, 06Z 136.8, 12Z 174.1, 18Z 145.4, 06-07 00Z 153.1, 06Z 159.3, 12Z 163.5, 18Z 176.2, 06-08 00Z 154, 06Z 163.4, 12Z 160.5, 18Z 168.8
openweather	Wind speed (wind_speed)	06-05 18Z 2.514, 06-06 00Z 2.195, 06Z 1.636, 12Z 1.706, 18Z 2.441, 06-07 00Z 1.969, 06Z 1.796, 12Z 2.877, 18Z 2.716, 06-08 00Z 1.899, 06Z 2.159, 12Z 2.611, 18Z 2.86

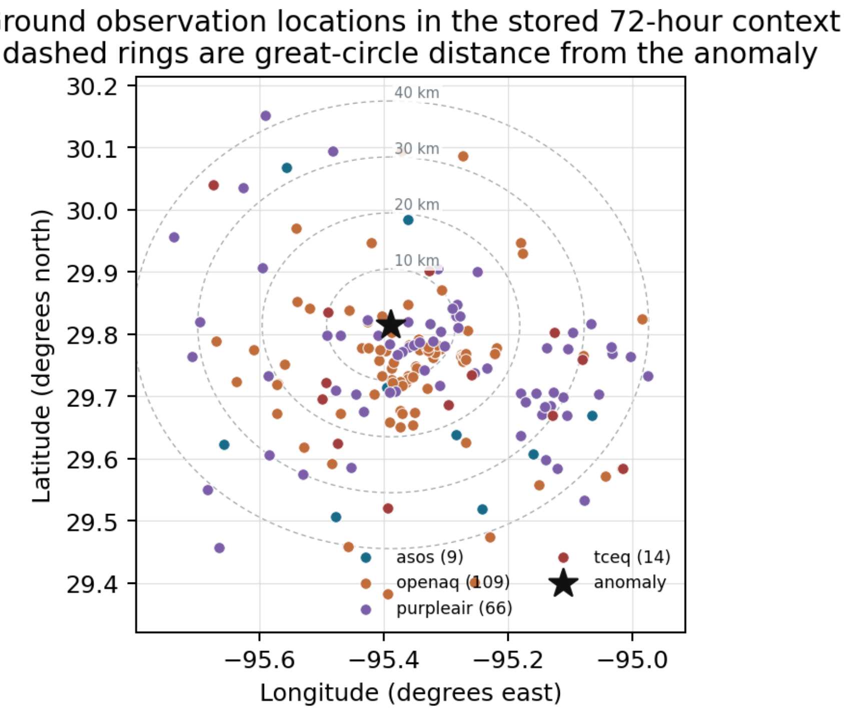
Source	Metric	Values
purpleair	PM2.5 (pm25)	06-05 18Z 2.464, 06-06 00Z 2.021, 06Z 1.871, 12Z 3.853, 18Z 3.907, 06-07 00Z 9.549, 06Z 10.39, 12Z 7.687, 18Z 6.841, 06-08 00Z 9.869, 06Z 10.17, 12Z 8.726, 18Z 6.713
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	CO granules available (s5p_co_granule_available)	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	NO2 granules available (s5p_no2_granule_available)	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	SO2 granules available (s5p_so2_granule_available)	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
tceq	Carbon monoxide (co)	06-05 18Z 0.2071, 06-06 00Z 0.2889, 06Z 0.2, 12Z 0.2722, 18Z 0.2778, 06-07 00Z 0.2778, 06Z 0.1333, 12Z 0.1722, 18Z 0.1933, 06-08 00Z 0.2389, 06Z 0.1667, 12Z 0.2611, 18Z 0.2167
tceq	Nitrogen dioxide (no2)	06-05 18Z 5.411, 06-06 00Z 5.392, 06Z 5.67, 12Z 3.539, 18Z 5.16, 06-07 00Z 3.372, 06Z 1.548, 12Z 1.372, 18Z 1.385, 06-08 00Z 2.668, 06Z 2.06, 12Z 3.814, 18Z 2.869
tceq	Sulfur dioxide (so2)	06-05 18Z -0.06667, 06-06 00Z -0.05, 06Z -0.1333, 12Z -0.1056, 18Z 0.04444, 06-07 00Z -0.1714, 06Z -0.2375, 12Z -0.2278, 18Z -0.1833, 06-08 00Z -0.1722, 06Z -0.25, 12Z -0.1588, 18Z -0.26

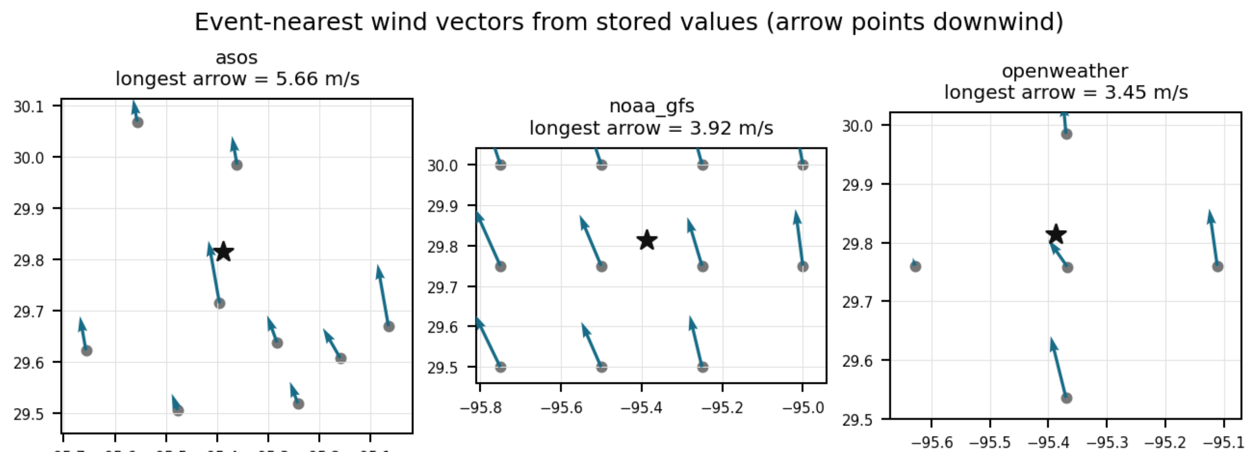
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	Relative humidity (humidity)	MCJ (11.201 km) 88.09, IAH (19.067 km) 85.44, HOU (22.155 km) 84.18, EFD (31.938 km) 86.22, DWH (32.527 km) 84.12, SGR (33.627 km) 86.99, T41 (35.174 km) 84.24, AXH (35.363 km) 81.19, +1 more
asos	Air temperature (temperature)	MCJ (11.201 km) 27.04, IAH (19.067 km) 27.21, HOU (22.155 km) 27.52, EFD (31.938 km) 27.62, DWH (32.527 km) 27.08, SGR (33.627 km) 27.29, T41 (35.174 km) 27.69, AXH (35.363 km) 27.39, +1 more
asos	Wind direction (wind_direction)	MCJ (11.201 km) 150.7, IAH (19.067 km) 143.5, HOU (22.155 km) 160.3, EFD (31.938 km) 142.9, DWH (32.527 km) 121, SGR (33.627 km) 145.8, T41 (35.174 km) 162.1, AXH (35.363 km) 117.9, +1 more
asos	Wind speed (wind_speed)	MCJ (11.201 km) 5.449, IAH (19.067 km) 3.601, HOU (22.155 km) 3.987, EFD (31.938 km) 4.519, DWH (32.527 km) 2.186, SGR (33.627 km) 3.815, T41 (35.174 km) 4.609, AXH (35.363 km) 1.851, +1 more
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.50 (12.999 km) 5873, gfs:29.75,-95.25 (15.102 km) 5874, gfs:30.00,-95.50 (23.292 km) 5872, gfs:30.00,-95.25 (24.525 km) 5873, gfs:29.75,-95.75 (35.695 km) 5872, gfs:29.50,-95.50 (36.619 km) 5874, gfs:29.50,-95.25 (37.419 km) 5875, gfs:29.75,-95.00 (38.097 km) 5876, +3 more
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.50 (12.999 km) 734.4, gfs:29.75,-95.25 (15.102 km) 726.6, gfs:30.00,-95.50 (23.292 km) 774.6, gfs:30.00,-95.25 (24.525 km) 725.4, gfs:29.75,-95.75 (35.695 km) 616.5, gfs:29.50,-95.50 (36.619 km) 569.3, gfs:29.50,-95.25 (37.419 km) 608.9, gfs:29.75,-95.00 (38.097 km) 642.6, +3 more
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.50 (12.999 km) 55.37, gfs:29.75,-95.25 (15.102 km) 55.45, gfs:30.00,-95.50 (23.292 km) 55, gfs:30.00,-95.25 (24.525 km) 55.21, gfs:29.75,-95.75 (35.695 km) 55.12, gfs:29.50,-95.50 (36.619 km) 55.58, gfs:29.50,-95.25 (37.419 km) 55.48, gfs:29.75,-95.00 (38.097 km) 55.31, +3 more

Source	Metric	Values
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.50 (12.999 km) 1008, gfs:29.75,-95.25 (15.102 km) 1009, gfs:30.00,-95.50 (23.292 km) 1006, gfs:30.00,-95.25 (24.525 km) 1008, gfs:29.75,-95.75 (35.695 km) 1007, gfs:29.50,-95.50 (36.619 km) 1009, gfs:29.50,-95.25 (37.419 km) 1010, gfs:29.75,-95.00 (38.097 km) 1010, +3 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.50 (12.999 km) 18.63, gfs:29.75,-95.25 (15.102 km) 18.53, gfs:30.00,-95.50 (23.292 km) 18.65, gfs:30.00,-95.25 (24.525 km) 18.54, gfs:29.75,-95.75 (35.695 km) 18.64, gfs:29.50,-95.50 (36.619 km) 18.64, gfs:29.50,-95.25 (37.419 km) 18.75, gfs:29.75,-95.00 (38.097 km) 18.59, +3 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.50 (12.999 km) -1.092, gfs:29.75,-95.25 (15.102 km) -1.103, gfs:30.00,-95.50 (23.292 km) -1.034, gfs:30.00,-95.25 (24.525 km) -1.075, gfs:29.75,-95.75 (35.695 km) -1.293, gfs:29.50,-95.50 (36.619 km) -1.376, gfs:29.50,-95.25 (37.419 km) -1.388, gfs:29.75,-95.00 (38.097 km) -1.201, +3 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.50 (12.999 km) 3.051, gfs:29.75,-95.25 (15.102 km) 3.053, gfs:30.00,-95.50 (23.292 km) 2.945, gfs:30.00,-95.25 (24.525 km) 2.815, gfs:29.75,-95.75 (35.695 km) 3.381, gfs:29.50,-95.50 (36.619 km) 3.131, gfs:29.50,-95.25 (37.419 km) 3.402, gfs:29.75,-95.00 (38.097 km) 3.37, +3 more
openaq	Ozone (ozone)	1319333 (4.693 km) 0.01897, 319 (10.034 km) 0.01549, 2046 (10.097 km) 0.01295, 310 (11.311 km) 0.0174, 3378 (15.436 km) 0.01613, 325 (16.839 km) 0.01622, 2039 (16.93 km) 0.01511, 311 (17.034 km) 0.01771, +10 more
openaq	PM10 (pm10)	15852419 (8.427 km) 41.4, 13380082 (15.436 km) 30.64, 271 (29.738 km) 28.61
openaq	PM2.5 (pm25)	2071327 (0.0 km) 12.35, 10107877 (1.388 km) 7.438, 13787197 (2.124 km) 7.37, 8967015 (3.659 km) 5.172, 12296914 (4.519 km) 8.864, 13787194 (4.585 km) 7.375, 8967373 (4.84 km) 7.417, 8967193 (4.94 km) 7.148, +80 more
openweather	Cloud cover (cloud_cover)	grid:center (6.48 km) 99.39, grid:north (19.034 km) 99.32, grid:west (24.006 km) 99.12, grid:east (27.362 km) 97.42, grid:south (31.042 km) 97.22
openweather	Relative humidity (humidity)	grid:center (6.48 km) 85.93, grid:north (19.034 km) 86.18, grid:west (24.006 km) 85.69, grid:east (27.362 km) 88.97, grid:south (31.042 km) 84.61
openweather	Precipitation rate (precipitation)	grid:center (6.48 km) 0.04986, grid:north (19.034 km) 0.0125, grid:west (24.006 km) 0.03917, grid:east (27.362 km) 0.02944, grid:south (31.042 km) 0.1351
openweather	Surface pressure (pressure)	grid:center (6.48 km) 1011, grid:north (19.034 km) 1011, grid:west (24.006 km) 1011, grid:east (27.362 km) 1011, grid:south (31.042 km) 1011
openweather	Air temperature (temperature)	grid:center (6.48 km) 27.68, grid:north (19.034 km) 27.43, grid:west (24.006 km) 27.56, grid:east (27.362 km) 27.41, grid:south (31.042 km) 27.69
openweather	Wind direction (wind_direction)	grid:center (6.48 km) 149.4, grid:north (19.034 km) 132.8, grid:west (24.006 km) 153.8, grid:east (27.362 km) 160.3, grid:south (31.042 km) 166.1
openweather	Wind speed (wind_speed)	grid:center (6.48 km) 2.123, grid:north (19.034 km) 1.964, grid:west (24.006 km) 1.46, grid:east (27.362 km) 3.584, grid:south (31.042 km) 1.944
purpleair	PM2.5 (pm25)	246011 (2.65 km) 5.299, 223813 (2.654 km) 11.2, 302037 (3.453 km) 8.34, 155367 (3.78 km) 7.766, 127451 (4.849 km) 2.175, 186303 (5.082 km) 7.384, 27821 (5.158 km) 0.3014, 296101 (5.395 km) 7.885, +58 more
tceq	Carbon monoxide (co)	482011052 (0.02 km) 0.3931, 482011035 (15.439 km) 0.1423, 482011039 (29.754 km) 0.1375
tceq	Nitrogen dioxide (no2)	482011052 (0.02 km) 7.085, 482010047 (10.026 km) 4.72, 482010024 (11.303 km) 0.6117, 482011066 (14.47 km) 4.326, 482011035 (15.439 km) 5.48, 482010416 (16.848 km) 3.825, 482010055 (17.04 km) 2.368, 482010026 (25.33 km) 6.054, +5 more
tceq	Sulfur dioxide (so2)	482011035 (15.439 km) -0.2264, 482010051 (22.781 km) -0.2662, 482011039 (29.754 km) 0.06765


Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly
40 km
30 km
20 km
10 km
Latitude (degrees north)
Longitude (degrees east)
30.2
30.1
30.0
29.9
29.8
29.7
29.6
29.5
29.4
−95.6
−95.4
−95.2
−95.0
asos (9)
openaq (109)
purpleair (66)
tceq (14)
anomaly


Event-nearest wind vectors from stored values (arrow points downwind)
asos
longest arrow = 5.66 m/s
noaa_gfs
longest arrow = 3.92 m/s
openweather
longest arrow = 3.45 m/s
30.1
30.0
29.9
29.8
29.7
29.6
29.5
−95.7
−95.6
−95.5
−95.4
−95.3
−95.2
−95.1
−95.8
−95.0

Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide
Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.
Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.
Missing or thin evidence means Unsure, not Invalid.
A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.
Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

The contemporaneous pollutant mix favors a non-combustion or weak-combustion particle source over a major fresh combustion plume because ozone was 0.011 ppm, nearby TCEQ CO was 0.2 ppm, nearby TCEQ NO2 was 2.0 ppb, and nearby TCEQ SO2 was about -0.2 ppb at 07:00 UTC.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 2 of 36

The openaq PM2.5 measurement at 29.815, -95.388 in Houston was 62.2 ug/m3 at 2026-06-07T07:00:00+00:00, compared with an expected value of 8.741958041958041 ug/m3, corresponding to a z-score of 10.913526865502508 and a severe anomaly classification.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 3 of 36

A severe thunderstorm or tornado event occurred in the area, stirring up pollutants and reducing air quality.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 4 of 36

The air quality anomaly in Houston, Texas on June 7th, 2026 is likely caused by a combination of factors including weather patterns and local emissions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 5 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 6 of 36

The air quality anomaly in Houston, Texas is caused by high levels of CO concentrations.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 7 of 36

An openaq sensor located at (29.815, -95.388) in Houston recorded a PM2.5 concentration spike to 62.2 ug/m3 at 2026-06-07T07:00:00+00:00, compared to an expected baseline of 8.74 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 8 of 36

Meteorological conditions were favorable for persistence and apparent enhancement of particulate matter because relative humidity reached 100 percent at the nearest ASOS observation near the anomaly time under cloudy nighttime conditions with modest southerly flow and a GFS planetary boundary layer height near 521 m.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 9 of 36

A coarse-plus-fine mechanically generated aerosol source such as road dust, construction activity, or other local resuspension mixed with humid aerosol is a plausible cause of the event because PM10 was simultaneously elevated while ozone, NO2, CO, and SO2 remained low, which weakens the case for a broad secondary photochemical haze or a major fresh combustion plume as the sole explanation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 10 of 36

On June 7, 2026, at 07:00 UTC, OpenAQ recorded a severe PM2.5 concentration peak of 62.2 ug/m3 in Houston, Texas.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 11 of 36

Humid nocturnal aerosol enhancement is supported because relative humidity was extremely high near the anomaly, reaching 100 percent at the nearest asos station around 06:55 UTC and about 93 percent in openweather near 06:57 UTC, while nighttime boundary-layer heights in gfs were modest at about 431 to 521 m, conditions that favor particle accumulation and hygroscopic growth.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 12 of 36

Concurrently elevated PM10 concentrations reaching 61 ug/m3 and elevated localized carbon monoxide levels near the anomaly site support localized pollutant accumulation under light southerly winds.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 13 of 36

A low nocturnal planetary boundary layer height averaging 431 meters around 06:00 UTC restricted atmospheric vertical mixing, facilitating surface aerosol accumulation prior to the PM2.5 peak recorded by OpenAQ.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 14 of 36

NOAA GFS atmospheric model data indicated a shallow nocturnal boundary layer height of approximately 431 meters around 06:00 UTC on June 7, 2026, trapping urban emissions near the surface.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 15 of 36

The PM2.5 levels are elevated at 8.7 ug/m3, which is higher than the mean of 6.5099 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 16 of 36

Humid nighttime boundary-layer conditions are a plausible contributing cause of the elevated PM2.5 because relative humidity was near saturation around the event and the boundary layer was only moderately deep overnight, conditions that favor particle accumulation and hygroscopic growth or optical over-response for fine-particle sensors.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 17 of 36

PurpleAir PM2.5 sensors within the region showed an increase in 6-hour average concentrations, rising to 10.39 ug/m3 around 06:00 UTC on June 7, 2026, from pre-event 6-hour means below 4.0 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 18 of 36

At 2026-06-07T07:00:00+00:00, the nearest openaq ozone measurement was 0.011 ppm, the collocated tceq CO measurement was 0.2 ppm, and the nearest tceq NO2 measurement was 2.0 ppb.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 19 of 36

High ambient relative humidity exceeding 90 percent promoted secondary aerosol formation and hygroscopic growth of particulate matter during early morning hours.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 20 of 36

A localized primary particulate plume from nearby combustion or industrial activity is a plausible cause of the PM2.5 spike because the anomalous monitor reported 62.2 ug/m3 while nearby PM2.5 observations were much lower, indicating strong spatial heterogeneity consistent with a narrow source influence rather than a uniform regional episode.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 21 of 36

The air quality anomaly in Houston, Texas on June 7th, 2026 is also characterized by elevated levels of NO2.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 22 of 36

A major fresh combustion plume or regional photochemical episode is contradicted because contemporaneous ozone near the anomaly was low at 0.011 ppm in openaq, nearby tceq CO was only 0.2 ppm, nearby tceq NO2 was about 2.0 ppb, and sentinel5p showed no clear CO or SO2 enhancement in the closest available overpass, which weakens support for a large fire, industrial sulfur plume, or broad secondary pollution event.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 23 of 36

The anomaly occurred on June 7th at around 07:00:00+00:00, with the nearest sensor located approximately 2.65 km away.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 24 of 36

Regional openaq PM10 measurements elevated around the time of the anomaly, reaching a 6-hour mean of 51.06 ug/m3 at 06:00 UTC on June 7, 2026, with the nearest monitor recording 61.0 ug/m3 at 07:00 UTC.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 25 of 36

Near-saturated relative humidity levels reaching up to 100 percent at 06:55 UTC supported aerosol hygroscopic growth and mass enhancement in the absence of precipitation washout.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 26 of 36

The air quality anomaly in Houston, Texas on June 7th, 2026 is characterized by elevated levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 27 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO2 concentrations.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 28 of 36

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, leading to poor air mixing and increased pollutant concentrations.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 29 of 36

A shallow nocturnal planetary boundary layer and calm surface winds trapped localized particulate emissions near the ground in Houston.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 30 of 36

ASOS meteorological observations recorded surface relative humidity levels reaching 100 percent near the anomaly location around 07:00 UTC on June 7, 2026, facilitating aerosol hygroscopic growth.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 31 of 36

A localized particulate source affecting the anomalous central Houston PM2.5 monitor is supported because the anomalous openaq PM2.5 value reached 62.2 ug/m3 at 07:00 UTC while the nearest purpleair PM2.5 reading was only 8.7 ug/m3 at the same time and the area-wide 6 h means were much lower than the anomaly, which indicates strong spatial heterogeneity inconsistent with a uniformly regional event.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 32 of 36

Anthropogenic emissions from central Houston industrial and vehicular sources accumulated under stable surface atmospheric conditions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 33 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 34 of 36

The nearest openaq PM10 measurement was 61.0 ug/m3 at 2026-06-07T07:00:00+00:00, indicating elevated coarse-plus-fine particulate mass at the same time as the PM2.5 anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 35 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 36 of 36

The openaq PM2.5 anomaly at central Houston is most consistent with a localized particulate plume rather than a regionally uniform PM2.5 event because the anomalous monitor reported 62.2 ug/m3 while the 6-hour network mean was 9.41 ug/m3 and the nearest purpleair monitor measured only 8.7 ug/m3 at the same time.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.